Foundations for statistical inference - Sampling distributions

In this lab, you will investigate the ways in which the statistics from a random sample of data can serve as point estimates for population parameters. We're interested in formulating a *sampling distribution* of our estimate in order to learn about the properties of the estimate, such as its distribution.

Setting a seed: We will take some random samples and build sampling distributions in this lab, which means you should set a seed at the start of your lab. If this concept is new to you, review the lab on probability.

Getting Started

Load packages

In this lab, we will explore and visualize the data using the **tidyverse** suite of packages. We will also use the **infer** package for resampling.

Let's load the packages.

```
library(tidyverse)
library(openintro)
library(infer)
```

The data

A 2019 Gallup report states the following:

> The premise that scientific progress benefits people has been embodied in discoveries throughout the ages – from the development of vaccinations to the explosion of technology in the past few decades, resulting in billions of supercomputers now resting in the hands and pockets of people worldwide. Still, not everyone around the world feels science benefits them personally.
>
> **Source:** World Science Day: Is Knowledge Power?

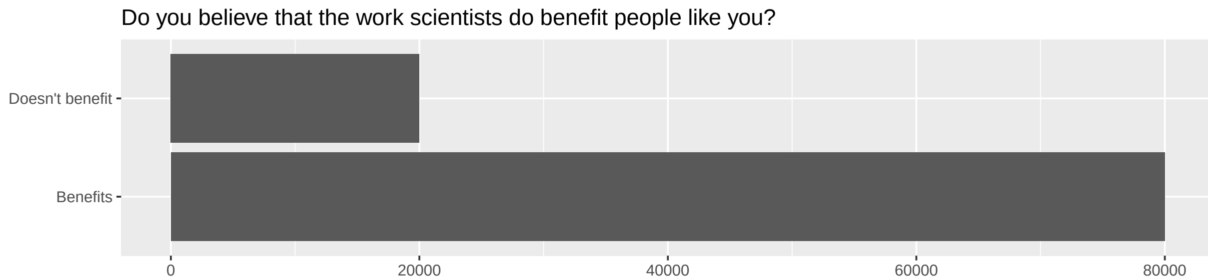
The Wellcome Global Monitor finds that 20% of people globally do not believe that the work scientists do benefits people like them. In this lab, you will assume this 20% is a true population proportion and learn about how sample proportions can vary from sample to sample by taking smaller samples from the population. We will first create our population assuming a population size of 100,000. This means 20,000 (20%) of the population think the work scientists do does not benefit them personally and the remaining 80,000 think it does.

```
global_monitor <- tibble(
  scientist_work = c(rep("Benefits", 80000), rep("Doesn't benefit", 20000))
)
```

The name of the data frame is `global_monitor` and the name of the variable that contains responses to the question *"Do you believe that the work scientists do benefit people like you?"* is `scientist_work`.

We can quickly visualize the distribution of these responses using a bar plot.

```r
ggplot(global_monitor, aes(x = scientist_work)) +
  geom_bar() +
  labs(
    x = "", y = "",
    title = "Do you believe that the work scientists do benefit people like you?"
  ) +
  coord_flip()
```



We can also obtain summary statistics to confirm we constructed the data frame correctly.

```r
global_monitor %>%
  count(scientist_work) %>%
  mutate(p = n /sum(n))
```

```
## # A tibble: 2 x 3
##   scientist_work      n     p
##   <chr>           <int> <dbl>
## 1 Benefits        80000   0.8
## 2 Doesn't benefit 20000   0.2
```

The unknown sampling distribution

In this lab, you have access to the entire population, but this is rarely the case in real life. Gathering information on an entire population is often extremely costly or impossible. Because of this, we often take a sample of the population and use that to understand the properties of the population.

If you are interested in estimating the proportion of people who don't think the work scientists do benefits them, you can use the `sample_n` command to survey the population.

```r
set.seed(1111)
samp1 <- global_monitor %>%
  sample_n(50)
```

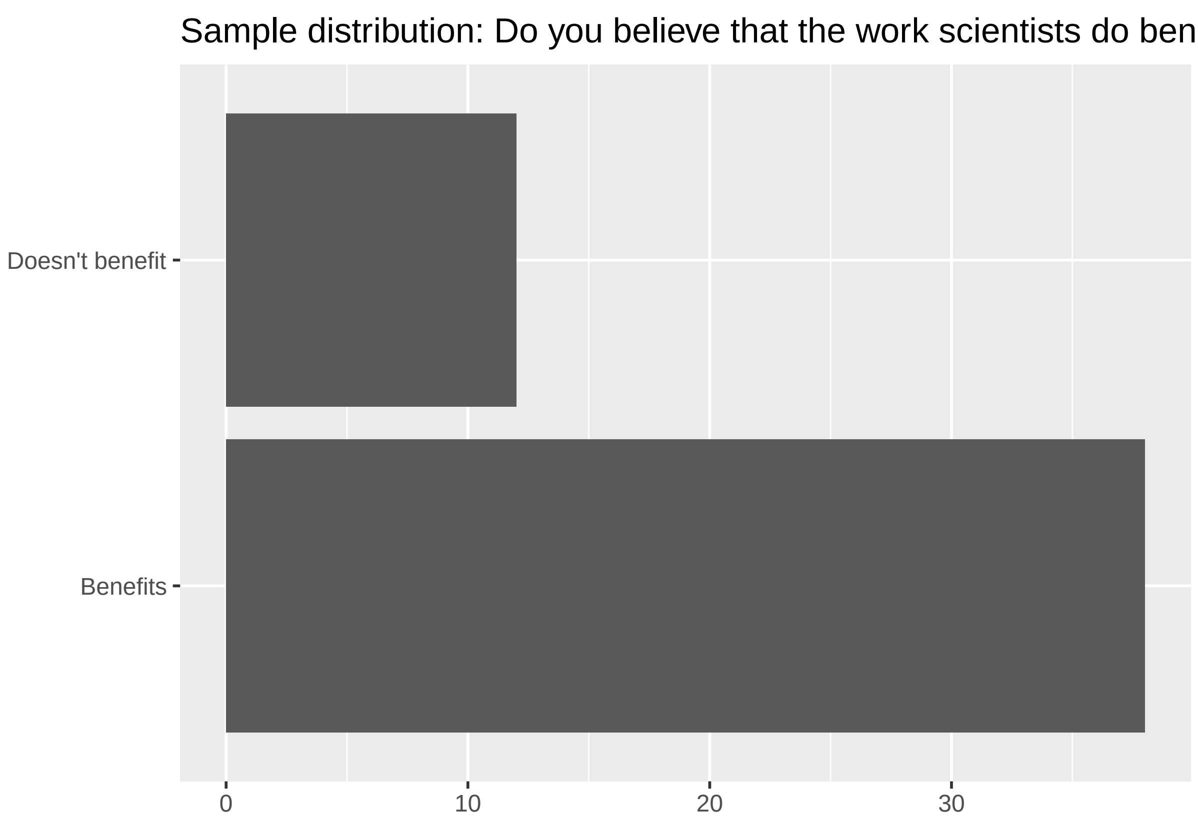
This command collects a simple random sample of size 50 from the `global_monitor` dataset, and assigns the result to `samp1`. This is similar to randomly drawing names from a hat that contains the names of all in the population. Working with these 50 names is considerably simpler than working with all 100,000 people in the population.

1. Describe the distribution of responses in this sample. How does it compare to the distribution of responses in the population. **Hint:** Although the `sample_n` function takes a random sample of observations (i.e. rows) from the dataset, you can still refer to the variables in the dataset with the same

names. Code you presented earlier for visualizing and summarizing the population data will still be useful for the sample, however be careful to not label your proportion `p` since you're now calculating a sample statistic, not a population parameters. You can customize the label of the statistics to indicate that it comes from the sample.

Step 1: Visualizing the sample distribution

```
# Plotting the sample distribution
ggplot(samp1, aes(x = scientist_work)) +
  geom_bar() +
  labs(
    x = "", y = "",
    title = "Sample distribution: Do you believe that the work scientists do benefit people like you?"
  ) +
  coord_flip()
```



Step 2: Summary Statistics of Sample distribution

```
samp1 %>%
  count(scientist_work) %>%
  mutate(sample_prop = n / sum(n))
```

```
## # A tibble: 2 x 3
##   scientist_work      n sample_prop
##   <chr>           <int>       <dbl>
```

```
## 1 Benefits             38         0.76
## 2 Doesn't benefit     12         0.24
```

In the sample of 50 people, we observe that:

76% of the sample (38 people) believe that the work scientists do benefits them. *24%* of the sample (12 people) believe that the work scientists do does not benefit them.

if we compare these proportions to the true population distribution:

The population has *80%* of people believing that the work scientists do benefits them and *20%* of people who think it does not benefit them.

By this we can say that,The sample is relatively close to the population distribution, but with a slight underestimation of those who think science benefits them i.e. 76% in the sample and 80% in the population and a slight overestimation of those who think it doesn't benefit them i.e. 24% in the sample and 20% in the population.

If you're interested in estimating the proportion of all people who do not believe that the work scientists do benefits them, but you do not have access to the population data, your best single guess is the sample mean.

```
samp1 %>%
  count(scientist_work) %>%
  mutate(p_hat = n /sum(n))
```

```
## # A tibble: 2 x 3
##   scientist_work      n p_hat
##   <chr>           <int> <dbl>
## 1 Benefits           38  0.76
## 2 Doesn't benefit    12  0.24
```

Depending on which 50 people you selected, your estimate could be a bit above or a bit below the true population proportion of 0.24. In general, though, the sample proportion turns out to be a pretty good estimate of the true population proportion, and you were able to get it by sampling less than 1% of the population.

2. Would you expect the sample proportion to match the sample proportion of another student's sample? Why, or why not? If the answer is no, would you expect the proportions to be somewhat different or very different? Ask a student team to confirm your answer.

No, I would not expect the sample proportion to match the sample proportion of another student's sample. This is because the sampling process is done by picking out the random numbers/variables from the proportion. Every time a sample is taken, there is a chance that the proportion of "Benefits" and "Doesn't benefit" responses could slightly differ due to the random selection of individuals from the population.

However, I would expect the proportions to be somewhat similar I mean it could be somewhat different, but not drastically so. if I run the same code over and over again without the seed I can clearly see that the sampling proportion is always different and no two sampling proportion actually completely matches

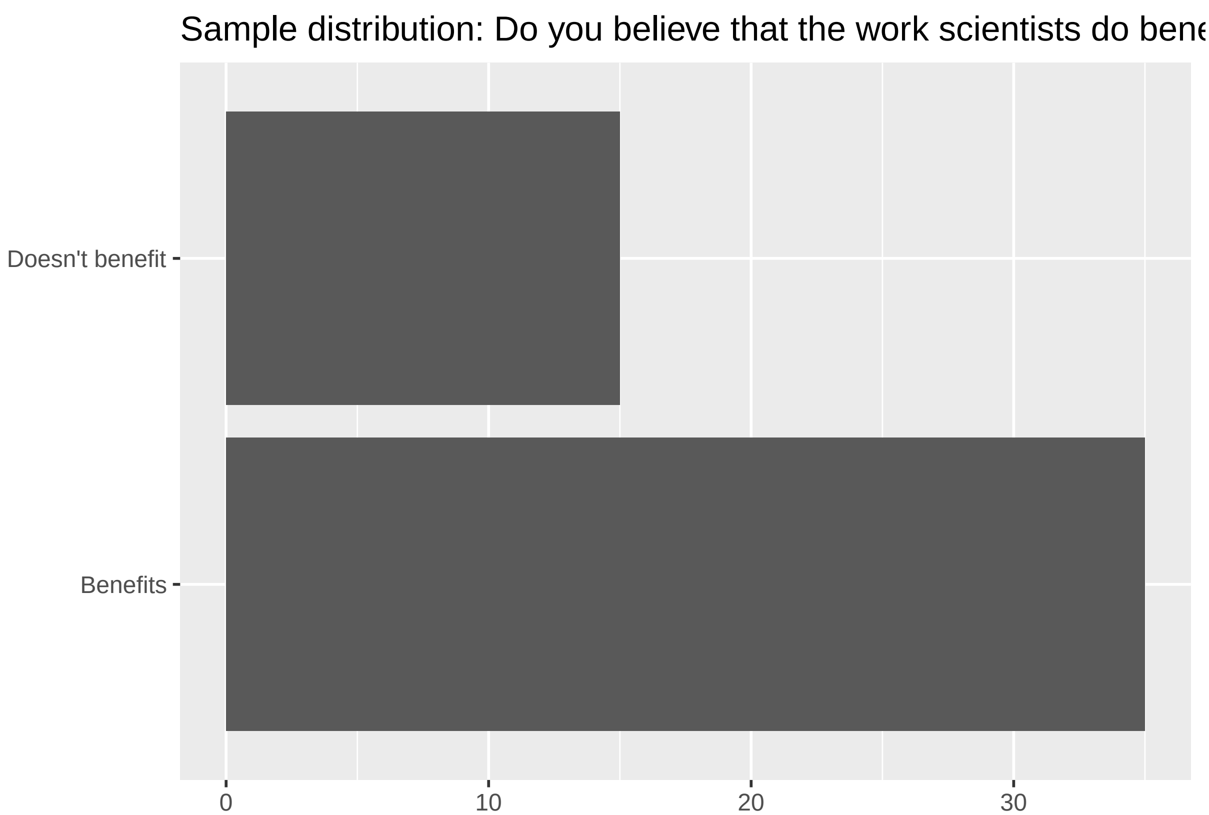
3. Take a second sample, also of size 50, and call it `samp2`. How does the sample proportion of `samp2` compare with that of `samp1`? Suppose we took two more samples, one of size 100 and one of size 1000. Which would you think would provide a more accurate estimate of the population proportion?

Step 1: initializing Samp2 of size 50

```
set.seed(1112)
samp2 <- global_monitor %>%
  sample_n(50)
```

Step 2: Visualizing the sample distribution

```
# Plotting the sample distribution
ggplot(samp2, aes(x = scientist_work)) +
  geom_bar() +
  labs(
    x = "", y = "",
    title = "Sample distribution: Do you believe that the work scientists do benefit people like you?"
  ) +
  coord_flip()
```



Step 3: Summary Statistics of Sample distribution

```
samp2 %>%
  count(scientist_work) %>%
  mutate(sample_prop = n / sum(n))
```

```
## # A tibble: 2 x 3
##   scientist_work      n sample_prop
```

```
##   <chr>           <int>       <dbl>
## 1 Benefits           35         0.7
## 2 Doesn't benefit    15         0.3
```

Lets compare the SAMP1 and SAMP2 Summary Statistics samp1:

Benefits: 76% (38 out of 50) Doesn't benefit: 24% (12 out of 50)

samp2:

Benefits: 70% (35 out of 50) Doesn't benefit: 30% (15 out of 50)

As, mentioned in the previous question the variables are picked up randomly and due to this there is aslight difference between both the sample proportions.

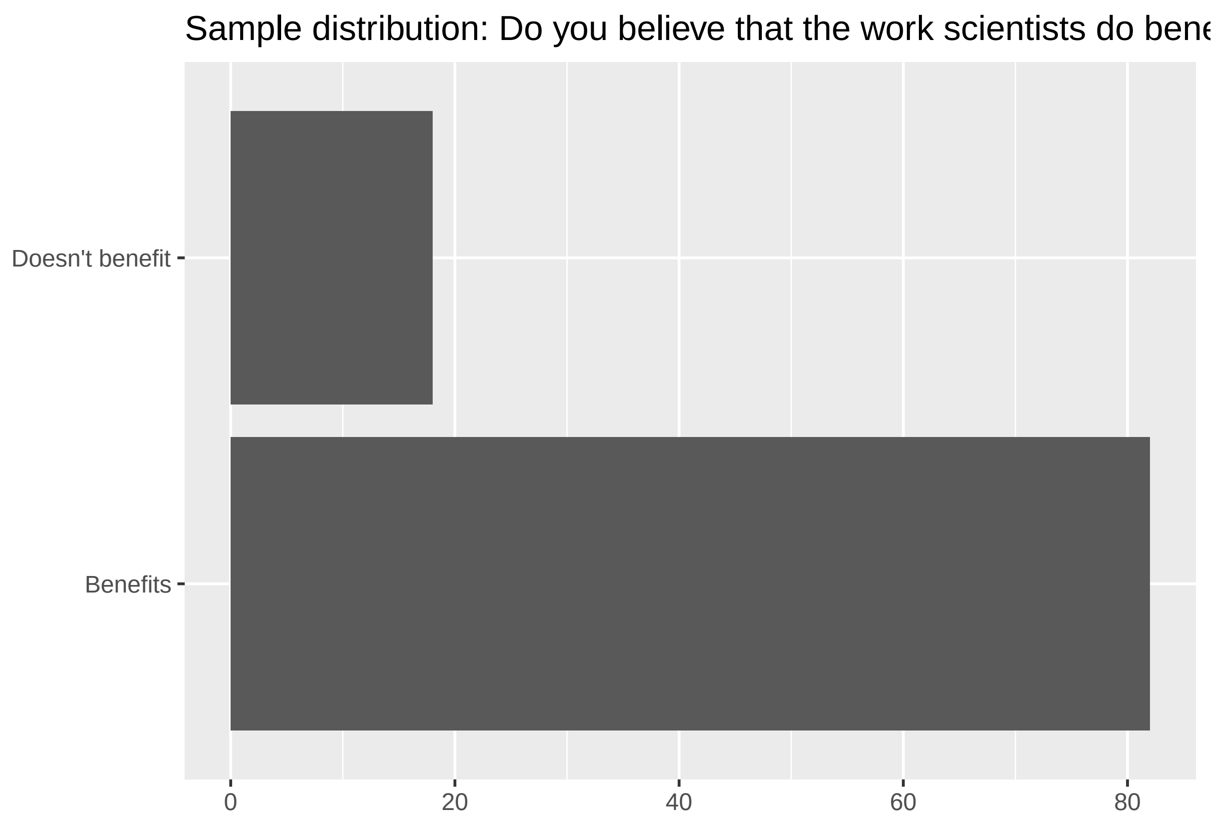
Drawing the samples of size 100 and 1000 respectively

Step 1: initializing Samp3 of size 100

```
set.seed(1113)
samp3 <- global_monitor %>%
  sample_n(100)
```

Step 2: Visualizing the sample distribution

```
# Plotting the sample distribution
ggplot(samp3, aes(x = scientist_work)) +
  geom_bar() +
  labs(
    x = "", y = "",
    title = "Sample distribution: Do you believe that the work scientists do benefit people like you?"
  ) +
  coord_flip()
```



Step 3: Summary Statistics of Sample distribution

```
samp3 %>%
  count(scientist_work) %>%
  mutate(sample_prop = n / sum(n))
```

```
## # A tibble: 2 x 3
##   scientist_work      n sample_prop
##   <chr>            <int>       <dbl>
## 1 Benefits            82        0.82
## 2 Doesn't benefit     18        0.18
```

Sample population of size 1000

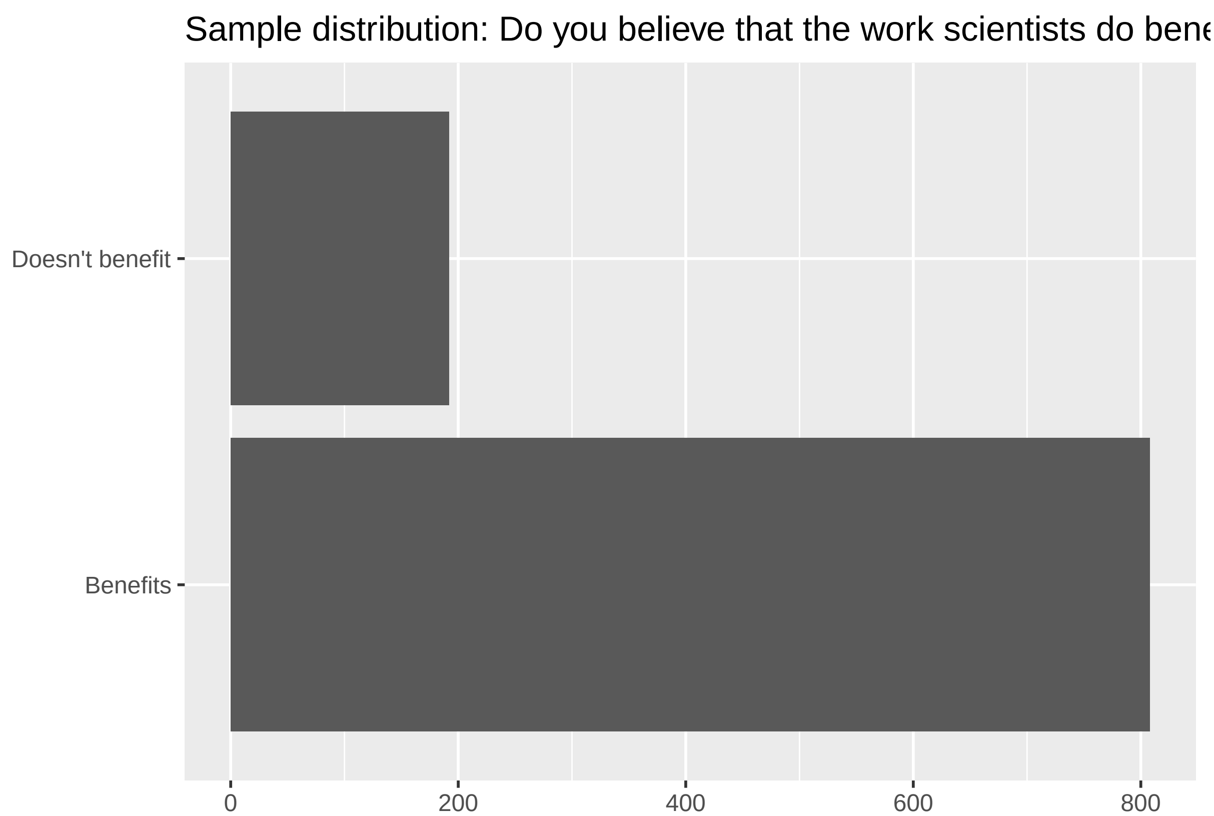
Step 1: initializing Samp4 of size 1000

```
set.seed(1114)
samp4 <- global_monitor %>%
  sample_n(1000)
```

Step 2: Visualizing the sample distribution

```
# Plotting the sample distribution
ggplot(samp4, aes(x = scientist_work)) +
  geom_bar() +
```

```
  labs(
    x = "", y = "",
    title = "Sample distribution: Do you believe that the work scientists do benefit people like you?"
  ) +
  coord_flip()
```



Step 3: Summary Statistics of Sample distribution

```
samp4 %>%
  count(scientist_work) %>%
  mutate(sample_prop = n / sum(n))
```

```
## # A tibble: 2 x 3
##   scientist_work      n sample_prop
##   <chr>           <int>       <dbl>
## 1 Benefits          808       0.808
## 2 Doesn't benefit   192       0.192
```

Lets compare SAMP3 and SAMP4 Summary Statistics

samp3 (size 100):

Benefits: 82% (82 out of 100) Doesn't benefit: 18% (18 out of 100)

samp4 (size 1000):

Benefits: 80.8% (808 out of 1000) Doesn't benefit: 19.2% (192 out of 1000)

It appears to be the same case with the samp1 and samp2 where there was the slight difference between the sample proportions in the same way, in the samp3 and samp4 there is also a slight difference in their proportions.

But it is clearer from these samples is that as the sample size increases the sample proportion is closer to the proportion of the population.

Not surprisingly, every time you take another random sample, you might get a different sample proportion. It's useful to get a sense of just how much variability you should expect when estimating the population mean this way. The distribution of sample proportions, called the *sampling distribution (of the proportion)*, can help you understand this variability. In this lab, because you have access to the population, you can build up the sampling distribution for the sample proportion by repeating the above steps many times. Here, we use R to take 15,000 different samples of size 50 from the population, calculate the proportion of responses in each sample, filter for only the *Doesn't benefit* responses, and store each result in a vector called `sample_props50`. Note that we specify that `replace = TRUE` since sampling distributions are constructed by sampling with replacement.

```
sample_props50 <- global_monitor %>%
                    rep_sample_n(size = 50, reps = 15000, replace = TRUE) %>%
                    count(scientist_work) %>%
                    mutate(p_hat = n /sum(n)) %>%
                    filter(scientist_work == "Doesn't benefit")
```

And we can visualize the distribution of these proportions with a histogram.

```
ggplot(data = sample_props50, aes(x = p_hat)) +
  geom_histogram(binwidth = 0.02) +
  labs(
    x = "p_hat (Doesn't benefit)",
    title = "Sampling distribution of p_hat",
    subtitle = "Sample size = 50, Number of samples = 15000"
  )
```

Next, you will review how this set of code works.

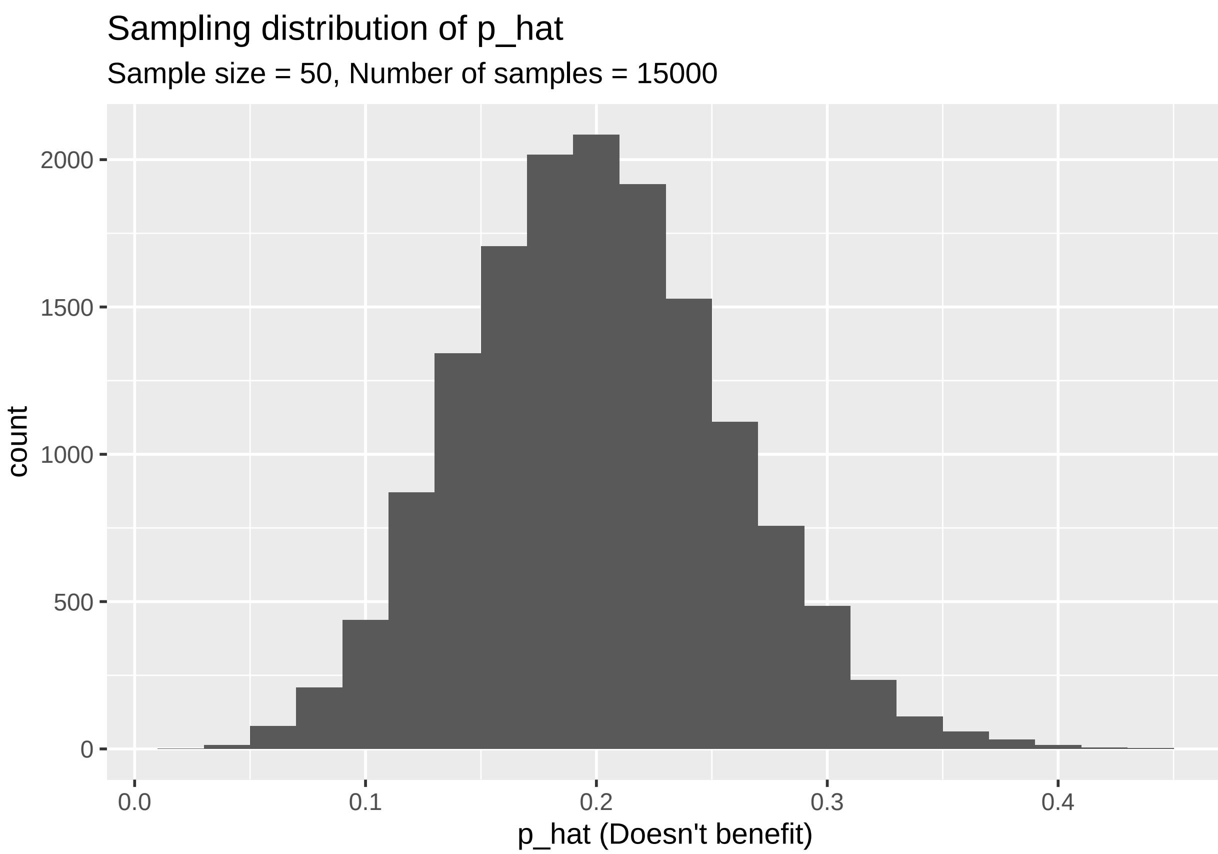
4. How many elements are there in `sample_props50`? Describe the sampling distribution, and be sure to specifically note its center. Make sure to include a plot of the distribution in your answer.

Number of elements in sample__props50: The sample__props50 was generated by taking 15,000 samples of size 50 from the population. And from this in the code that there's a filter applied on the question *Doesn't Benefit from the work of Scientist.* Then we can consider that there are *15,000 rows* of *Doesn't Benefit* in th sample__props50 data frame.

Description of the sampling distribution: The sampling distribution of p__hat, which represents the proportion of individuals who doesn't believe that science benefits them, is normal in shape, as shown in the histogram below. So, the distribution is of normal distribution. The mean in the distribution is around 0.2, which actually matches the true population proportion i.e. 20% of people who believe that the work scientists do doesn't benefit them.

By this we can tell that the mean of the sampling distribution is almost similar or close to the true population proportion of the Doesn't benefit data set. And the distribution is a normal distribution.

```
ggplot(data = sample_props50, aes(x = p_hat)) +
  geom_histogram(binwidth = 0.02) +
  labs(
    x = "p_hat (Doesn't benefit)",
    title = "Sampling distribution of p_hat",
    subtitle = "Sample size = 50, Number of samples = 15000"
  )
```



Interlude: Sampling distributions

The idea behind the `rep_sample_n` function is *repetition*. Earlier, you took a single sample of size `n` (50) from the population of all people in the population. With this new function, you can repeat this sampling procedure `rep` times in order to build a distribution of a series of sample statistics, which is called the **sampling distribution**.

Note that in practice one rarely gets to build true sampling distributions, because one rarely has access to data from the entire population.

Without the `rep_sample_n` function, this would be painful. We would have to manually run the following code 15,000 times

```
global_monitor %>%
  sample_n(size = 50, replace = TRUE) %>%
  count(scientist_work) %>%
  mutate(p_hat = n /sum(n)) %>%
  filter(scientist_work == "Doesn't benefit")
```

```
## # A tibble: 1 x 3
##   scientist_work      n p_hat
##   <chr>           <int> <dbl>
## 1 Doesn't benefit    13  0.26
```

as well as store the resulting sample proportions each time in a separate vector.

Note that for each of the 15,000 times we computed a proportion, we did so from a **different** sample!

5. To make sure you understand how sampling distributions are built, and exactly what the `rep_sample_n` function does, try modifying the code to create a sampling distribution of **25 sample proportions** from **samples of size 10**, and put them in a data frame named `sample_props_small`. Print the output. How many observations are there in this object called `sample_props_small`? What does each observation represent?

```r
sample_props_small <- global_monitor %>%
                    rep_sample_n(size = 10, reps = 25, replace = TRUE) %>%
                    count(scientist_work) %>%
                    mutate(p_hat = n /sum(n)) %>%
                    filter(scientist_work == "Doesn't benefit")
print(sample_props_small)
```

```
## # A tibble: 22 x 4
## # Groups:   replicate [22]
##    replicate scientist_work      n p_hat
##        <int> <chr>           <int> <dbl>
##  1         1 Doesn't benefit     2   0.2
##  2         2 Doesn't benefit     2   0.2
##  3         3 Doesn't benefit     1   0.1
##  4         4 Doesn't benefit     3   0.3
##  5         5 Doesn't benefit     4   0.4
##  6         6 Doesn't benefit     1   0.1
##  7         7 Doesn't benefit     4   0.4
##  8         8 Doesn't benefit     1   0.1
##  9        10 Doesn't benefit     2   0.2
## 10        11 Doesn't benefit     2   0.2
## # i 12 more rows
```

As per the code there should be 25 rows but we can see in the output only 22 rows this might be because that may be in some rows the no of people who says that the work scientist doesn't benefit is 0, that might be the reason why there are only 22 rows.

Each observation (row) in the sample_props_small data frame represents:

1. A specific replicate (sample).
2. The number of "Doesn't benefit" responses (n) in that replicate.
3. The calculated proportion of "Doesn't benefit" responses in that replicate (p_hat), which is n/10, since each sample has 10 observations.

Sample size and the sampling distribution

Mechanics aside, let's return to the reason we used the `rep_sample_n` function: to compute a sampling distribution, specifically, the sampling distribution of the proportions from samples of 50 people.

```r
ggplot(data = sample_props50, aes(x = p_hat)) +
  geom_histogram(binwidth = 0.02)
```

The sampling distribution that you computed tells you much about estimating the true proportion of people who think that the work scientists do doesn't benefit them. Because the sample proportion is an unbiased estimator, the sampling distribution is centered at the true population proportion, and the spread of the distribution indicates how much variability is incurred by sampling only 50 people at a time from the population.

In the remainder of this section, you will work on getting a sense of the effect that sample size has on your sampling distribution.

6. Use the app below to create sampling distributions of proportions of *Doesn't benefit* from samples of size 10, 50, and 100. Use 5,000 simulations. What does each observation in the sampling distribution represent? How does the mean, standard error, and shape of the sampling distribution change as the sample size increases? How (if at all) do these values change if you increase the number of simulations? (You do not need to include plots in your answer.)

Insert your answer here

Each observation in the sampling distribution represents the proportion (p_hat) of people in a sample who believe that the work scientists do "Doesn't benefit" them. With a sample size of 10,50 and 100 each observation is the proportion for one of the 5,000 samples.

As the sample size increases: Mean: The mean of the sampling distribution will stay close to the true population proportion (around 0.2), when i run stimulation with sample size 10 i got the mean somewhere around 0.22, with sample size 50 has the mean is around 0.2 and the sample size 100 has the mean around 0.2

By this we can say that the mean is very much close to the population propotion

Standard Error (SE): The SE is decreasing as the sample size increasing. For sample size of 10 the SE is around 0.11, for the sample size 50 the SE is around 0.06 and for the sample size of 100 the SE is around the 0.04 , this is indicating less variability.

Shape: The shape of the distribution is almost normal and tighter around the true mean as the sample size increases, due to the Central Limit Theorem.

As, we have observed in from the graphs that as the sample size increases the graphs becomes more smooth and the distribution looks more normal

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More Practice

So far, you have only focused on estimating the proportion of those you think the work scientists doesn't benefit them. Now, you'll try to estimate the proportion of those who think it does.

Note that while you might be able to answer some of these questions using the app, you are expected to write the required code and produce the necessary plots and summary statistics. You are welcome to use the app for exploration.

7. Take a sample of size 15 from the population and calculate the proportion of people in this sample who think the work scientists do enhances their lives. Using this sample, what is your best point estimate of the population proportion of people who think the work scientists do enhances their lives?

Insert your answer here

Step 1: Taking a sample of size 15

```
set.seed(1115)
samp15 <- global_monitor %>% sample_n(15)
```

Step 2: Calculating the proportion of people who think the work scientists do enhances their lives

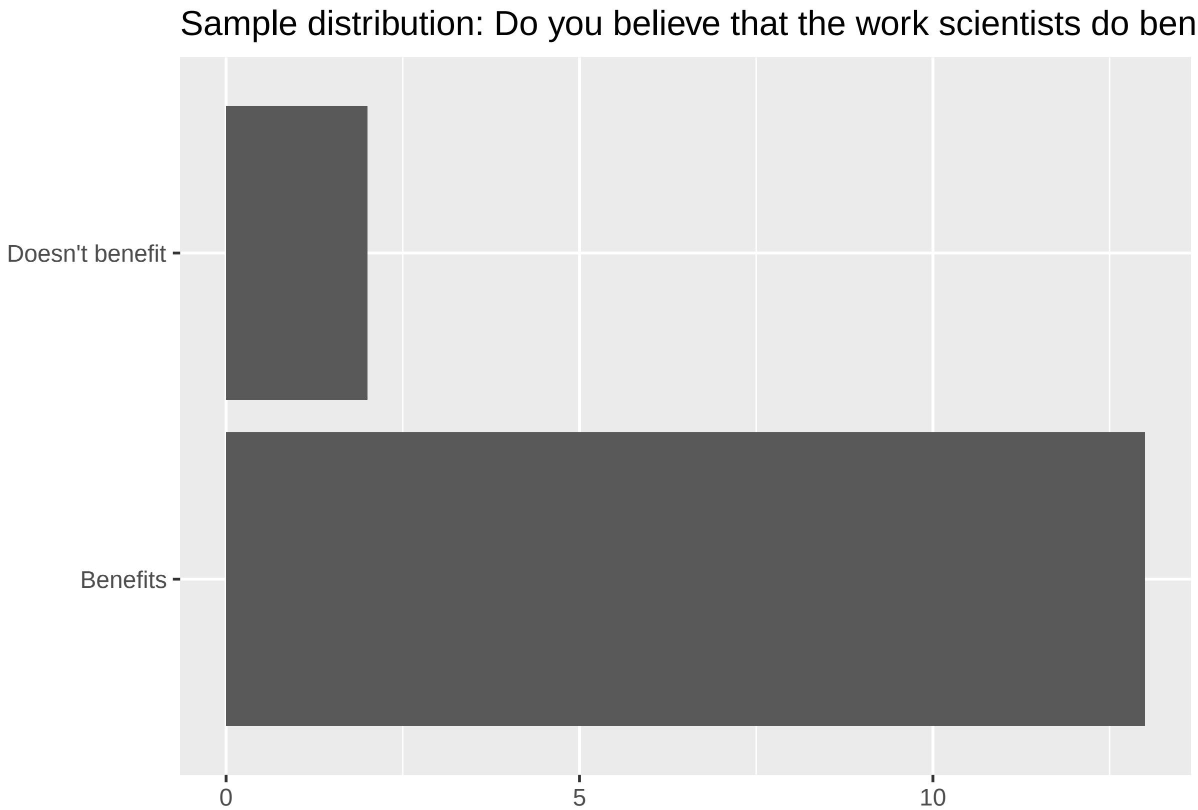
```
samp15_prop <- samp15 %>%
  count(scientist_work) %>%
  mutate(sample_prop = n / sum(n))

samp15_prop
```

```
## # A tibble: 2 x 3
##   scientist_work      n sample_prop
##   <chr>           <int>       <dbl>
## 1 Benefits           13       0.867
## 2 Doesn't benefit     2       0.133
```

Step 3: Plotting the sample distribution

```
ggplot(samp15, aes(x = scientist_work)) +
  geom_bar() +
  labs(
    x = "", y = "",
    title = "Sample distribution: Do you believe that the work scientists do benefits you?"
  ) +
  coord_flip()
```



In a sample of 15 people, 86.7% (13 out of 15) believe that the work scientists do benefits them, while 13.3% (2 out of 15) believe it does not.

The best point estimate for the population proportion of people who think scientists' work enhances their lives, based on this sample, is 86.7%.

The true population proportion i.e. 80% is close to this estimate, but the sample estimate is slightly higher due to the small sample size.

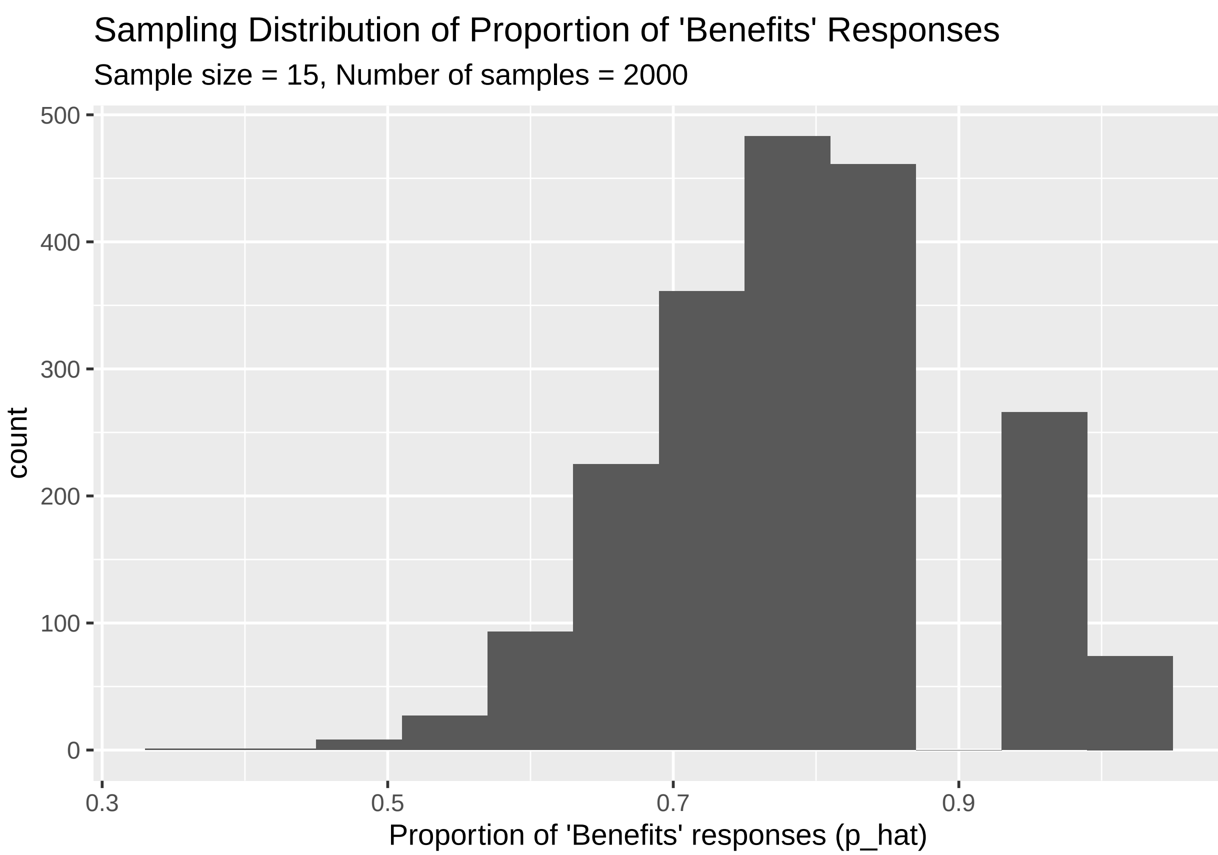
8. Since you have access to the population, simulate the sampling distribution of proportion of those who think the work scientists do enchances their lives for samples of size 15 by taking 2000 samples from the population of size 15 and computing 2000 sample proportions. Store these proportions in as `sample_props15`. Plot the data, then describe the shape of this sampling distribution. Based on this sampling distribution, what would you guess the true proportion of those who think the work scientists do enhances their lives to be? Finally, calculate and report the population proportion.

Step 1: Simulating the sampling distribution

```
set.seed(1116)
sample_props15 <- global_monitor %>%
  rep_sample_n(size = 15, reps = 2000, replace = TRUE) %>%
  count(scientist_work) %>%
  mutate(p_hat = n / sum(n)) %>%
  filter(scientist_work == "Benefits")
```

Step 2: Plotting the sampling distribution

```r
# Plotting the sampling distribution
ggplot(data = sample_props15, aes(x = p_hat)) +
  geom_histogram(binwidth = 0.06) +
  labs(
    x = "Proportion of 'Benefits' responses (p_hat)",
    title = "Sampling Distribution of Proportion of 'Benefits' Responses",
    subtitle = "Sample size = 15, Number of samples = 2000"
  )
```



1. Shape: The sampling distribution of sample proportions is approximately normal, centered around the population proportion of 0.8.
2. Estimated true proportion: Based on the sampling distribution, the true proportion of people who think the work scientists do enhances their lives is around 80%.

Step 3: Calculate and report the true population proportion

```r
true_prop <- global_monitor %>%
  count(scientist_work) %>%
  mutate(population_prop = n / sum(n)) %>%
  filter(scientist_work == "Benefits")

true_prop
```

```
## # A tibble: 1 x 3
##   scientist_work     n population_prop
##   <chr>          <int>           <dbl>
## 1 Benefits       80000             0.8
```

Actual population proportion: The actual population proportion is 80%, confirming that the sampling distribution gives a good estimate of the population value.

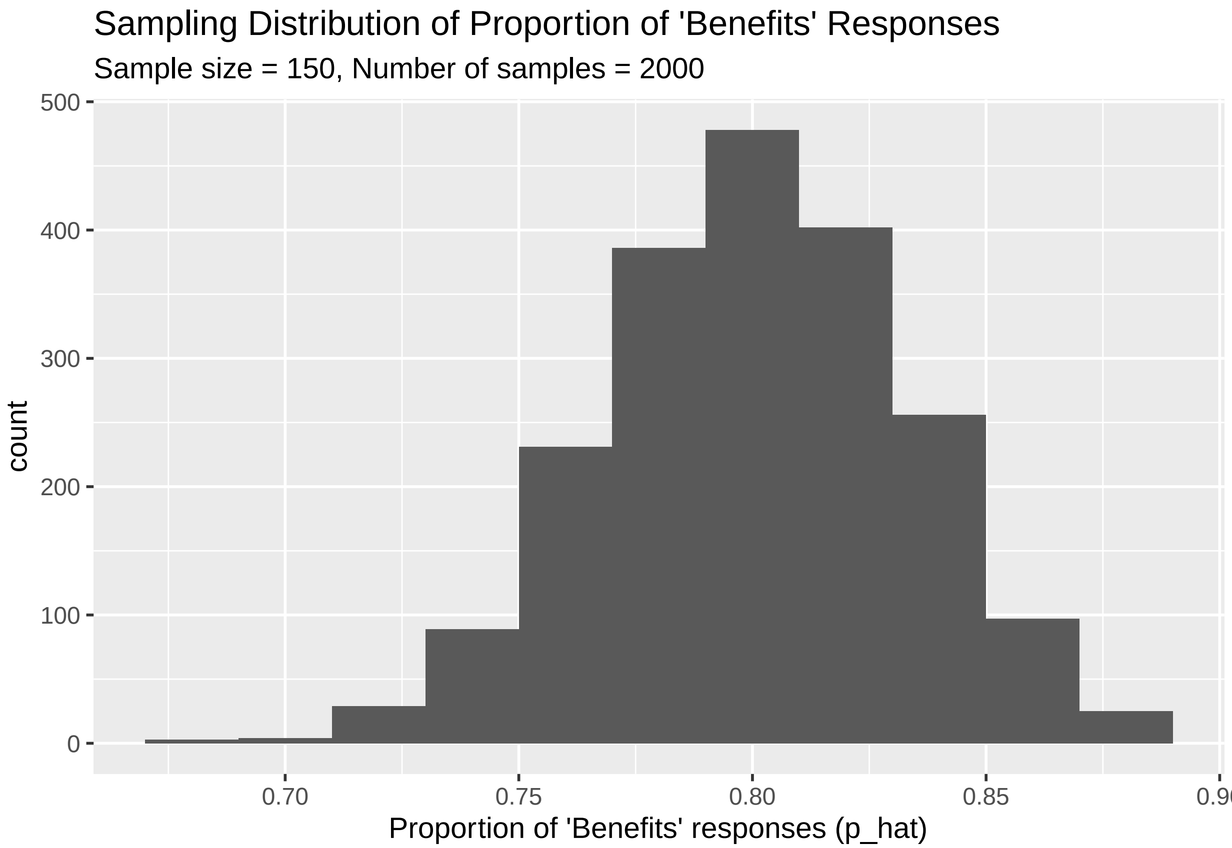
9. Change your sample size from 15 to 150, then compute the sampling distribution using the same method as above, and store these proportions in a new object called `sample_props150`. Describe the shape of this sampling distribution and compare it to the sampling distribution for a sample size of 15. Based on this sampling distribution, what would you guess to be the true proportion of those who think the work scientists do enhances their lives?

Step 1: Simulating the sampling distribution

```
set.seed(1117)
sample_props150 <- global_monitor %>%
  rep_sample_n(size = 150, reps = 2000, replace = TRUE) %>%
  count(scientist_work) %>%
  mutate(p_hat = n / sum(n)) %>%
  filter(scientist_work == "Benefits")
```

Step 2: Plotting the sampling distribution

```
# Plotting the sampling distribution
ggplot(data = sample_props150, aes(x = p_hat)) +
  geom_histogram(binwidth = 0.02) +
  labs(
    x = "Proportion of 'Benefits' responses (p_hat)",
    title = "Sampling Distribution of Proportion of 'Benefits' Responses",
    subtitle = "Sample size = 150, Number of samples = 2000"
  )
```



1. Shape: The sampling distribution of sample proportions is approximately normal, centered the population proportion of 0.8.
2. Estimated true proportion: Based on the sampling distribution, the true proportion of people who think the work scientists do enhances their lives is exactly near 80%.

Step 3: Calculate and report the true population proportion

```
true_prop_1 <- global_monitor %>%
  count(scientist_work) %>%
  mutate(population_prop = n / sum(n)) %>%
  filter(scientist_work == "Benefits")

true_prop_1
```

```
## # A tibble: 1 x 3
##   scientist_work     n population_prop
##   <chr>          <int>           <dbl>
## 1 Benefits       80000             0.8
```

Actual population proportion: The actual population proportion is 80%, confirming that the sampling distribution gives a exact estimate of the population value.

10. Of the sampling distributions from 2 and 3, which has a smaller spread? If you're concerned with making estimates that are more often close to the true value, would you prefer a sampling distribution with a large or small spread?

The sampling distribution from 2 has the smaller spread, if I want to make estimates that are more often close to the true value I would prefer a sampling distribution with a smaller spread and larger sample size. As, we all know that bigger the sample size smaller the spread.
